

Higher-order clustering for 3D point clouds

Exact density hierarchies, geometric priors and a foundation-model perspective

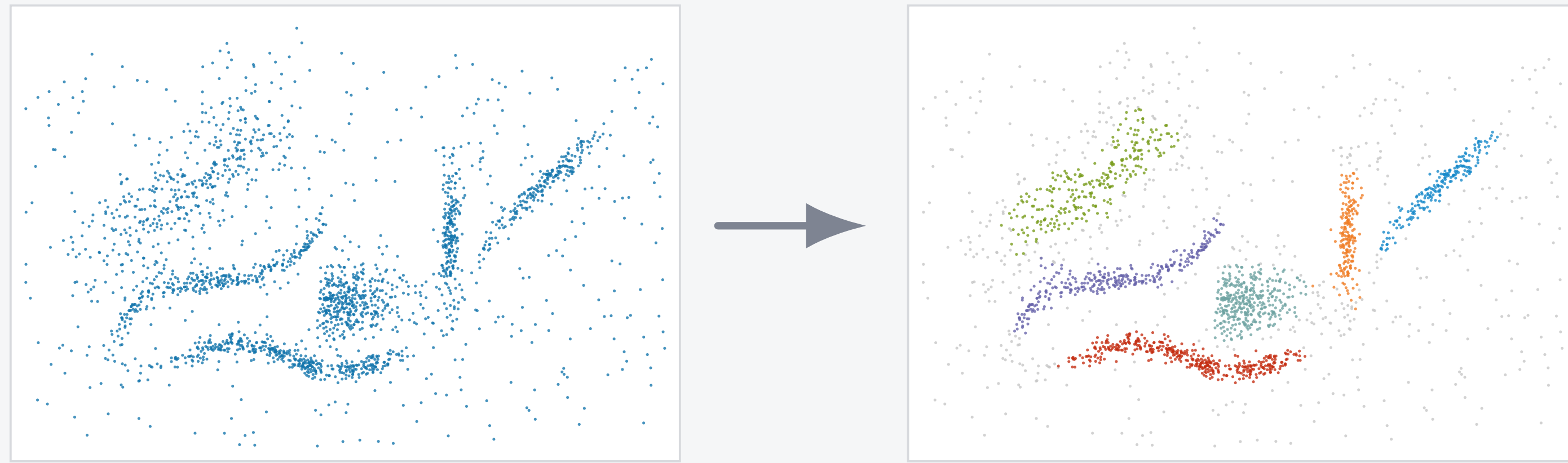
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1 What is clustering?

Group similar observations without class labels. Density-based clustering can recover non-convex groups and leave noisy points unassigned.

Unlabelled points

HDBSCAN [3]



2D example: HDBSCAN [3]. Colours indicate clusters; grey points remain unassigned.

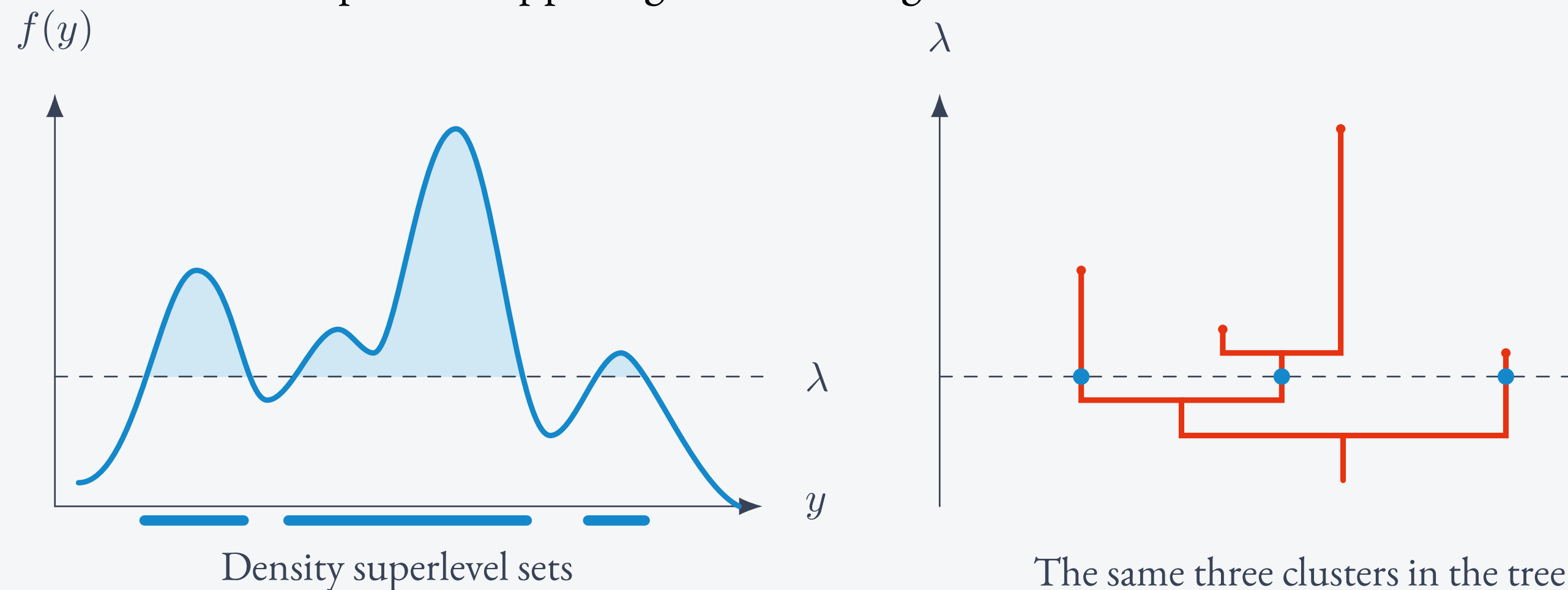
A partition gives *groups*; a hierarchy also tells us **at which scale they merge**.

2 The mathematical model

For a probability density f on $\mathbb{R}^d$, Hartigan's clusters are the **connected components** of a density superlevel set [1]:

$$E_\lambda = \{y \in \mathbb{R}^d : f(y) \geq \lambda\}.$$

As λ decreases, components appear, grow and merge.



For $X = \{x_1, \dots, x_n\}$, estimate density from the distance $r_K(y)$ to the K -th nearest sample point [2]:

$$\hat{f}_K(y) = \frac{K}{n \omega_d r_K(y)^d}.$$

ω_d is the volume of the unit ball; K controls the local density scale.

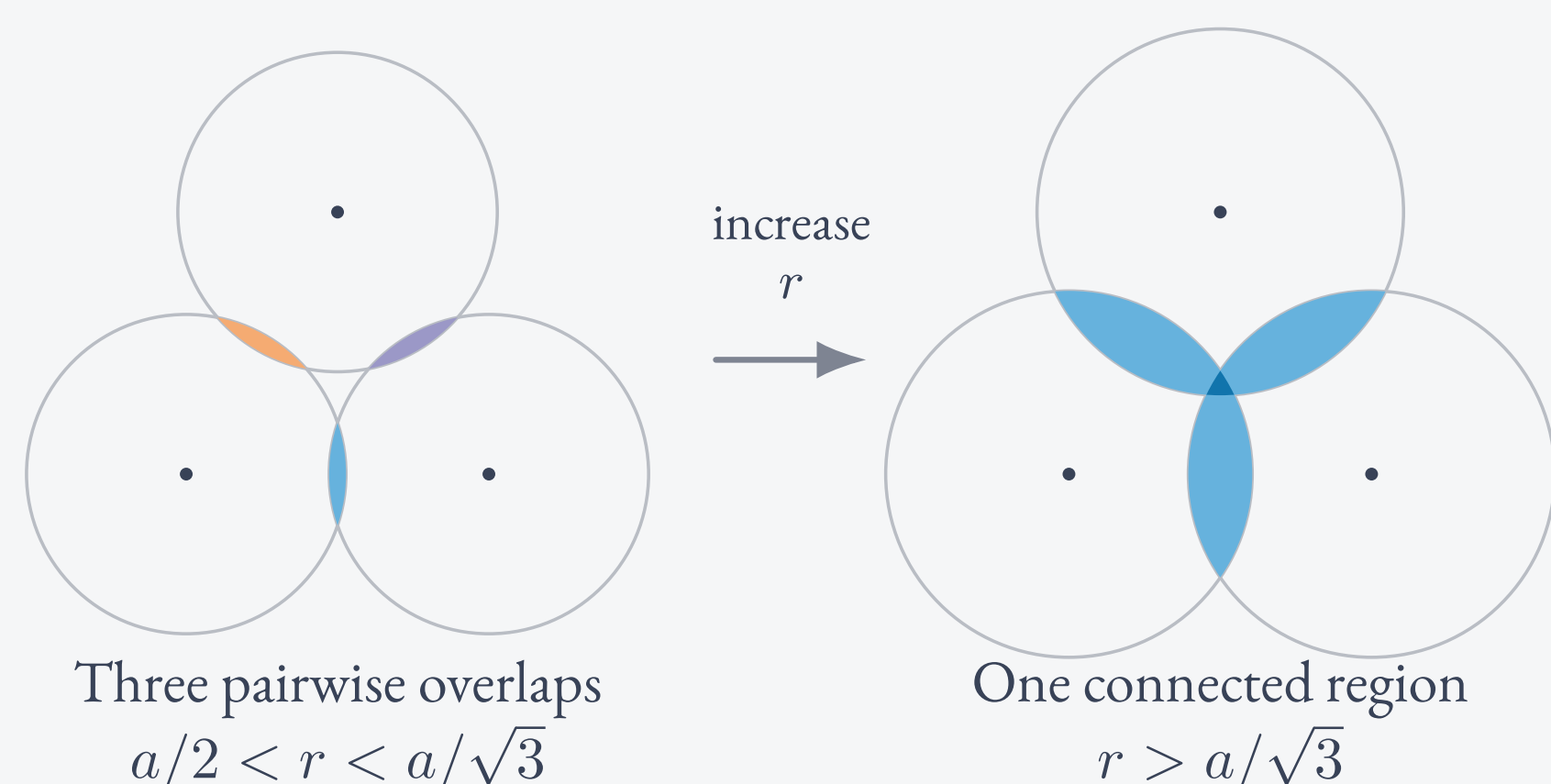
$$\text{High density} \iff \text{small neighbour radius} \\ \{\hat{f}_K \geq \lambda\} = \{r_K \leq r\}, \quad r = (K/(n\omega_d\lambda))^{1/d}.$$

3 HGP: connect at the right order

The region covered by at least K closed balls is

$$E_{K,r} = \{y : \#(X \cap B(y, r)) \geq K\} = \bigcup_{\substack{S \subseteq X \\ |S|=K}} \bigcap_{x \in S} B(x, r).$$

Replace edges by higher-order interactions. In the Čech complex, connect $(K-1)$ -simplices through K -simplices: their components are K -polyhedra [4,5].



$K=2$ on an equilateral triangle of side a : the three pairwise overlaps merge at $r = a/\sqrt{3}$. Connectivity is through triangles, not a shared vertex.

Exact correspondence. The K -polyhedra recover the connected components of $E_{K,r}$ at every radius [4].

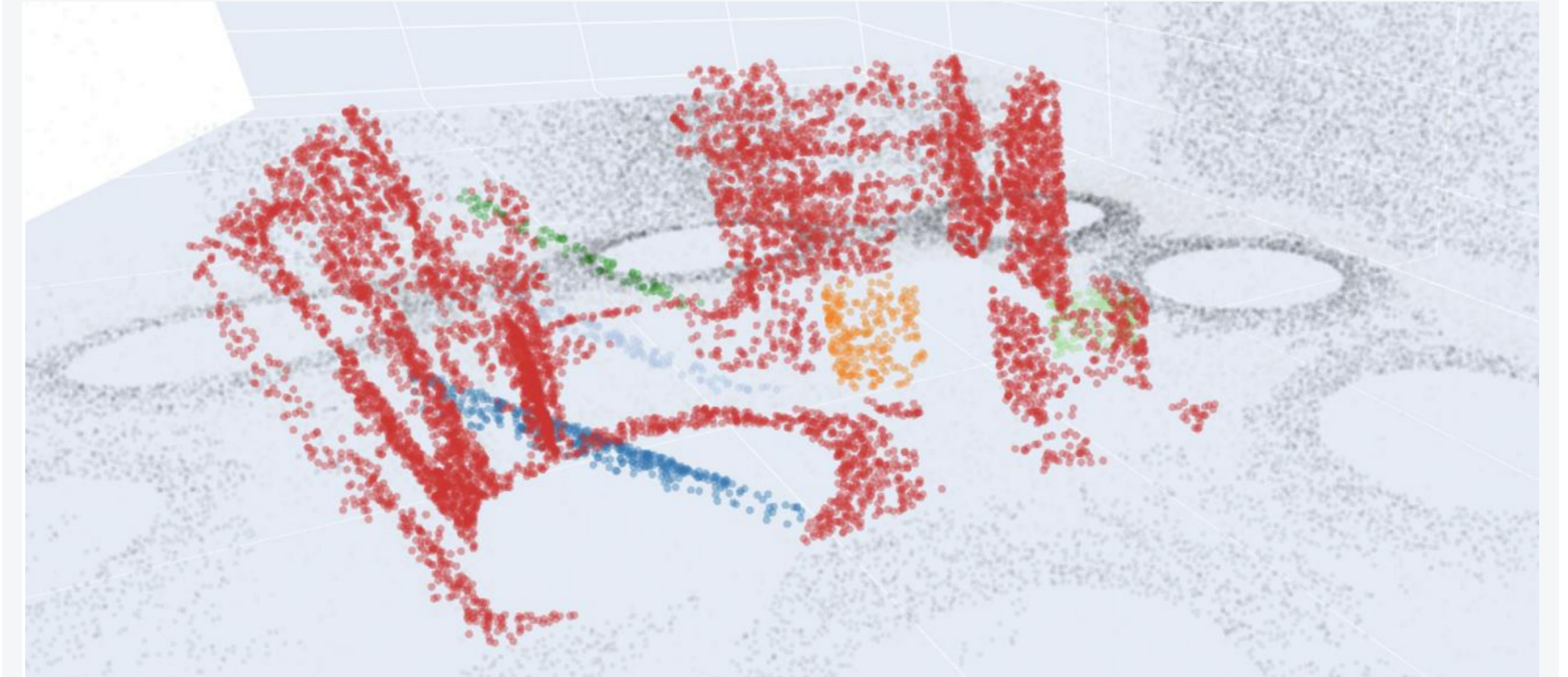
HGP-Clusterer uses geometric sparsification to process critical merges. $K=1$ gives Single-Linkage (minimum spanning tree); for $K \geq 2$, clusters may share points [4].

Selected references

- [1] J. A. Hartigan. *Consistency of single linkage for high-density clusters*. Journal of the American Statistical Association, 76:388–394, 1981.
- [2] G. Biau and L. Devroye. *Lectures on the Nearest Neighbor Method*. Springer, 2015.
- [3] R. J. G. B. Campello, D. Moulavi and J. Sander. *Density-Based Clustering Based on Hierarchical Density Estimates*. PAKDD, pp. 160–172, 2013.

4 LiDAR: find unexpected objects

Detect objects present in a LiDAR scan but absent from a reference 3D model. In collaboration with Marie ASPRO (Inria Startup Studio).



Synthetic shipyard benchmark: 2,000,000 points. Red: reference-consistent points. Other colours: detected anomalies. Grey: background. © Naval Group [6].

HGP
hierarchy

3D-model-guided
tree selection

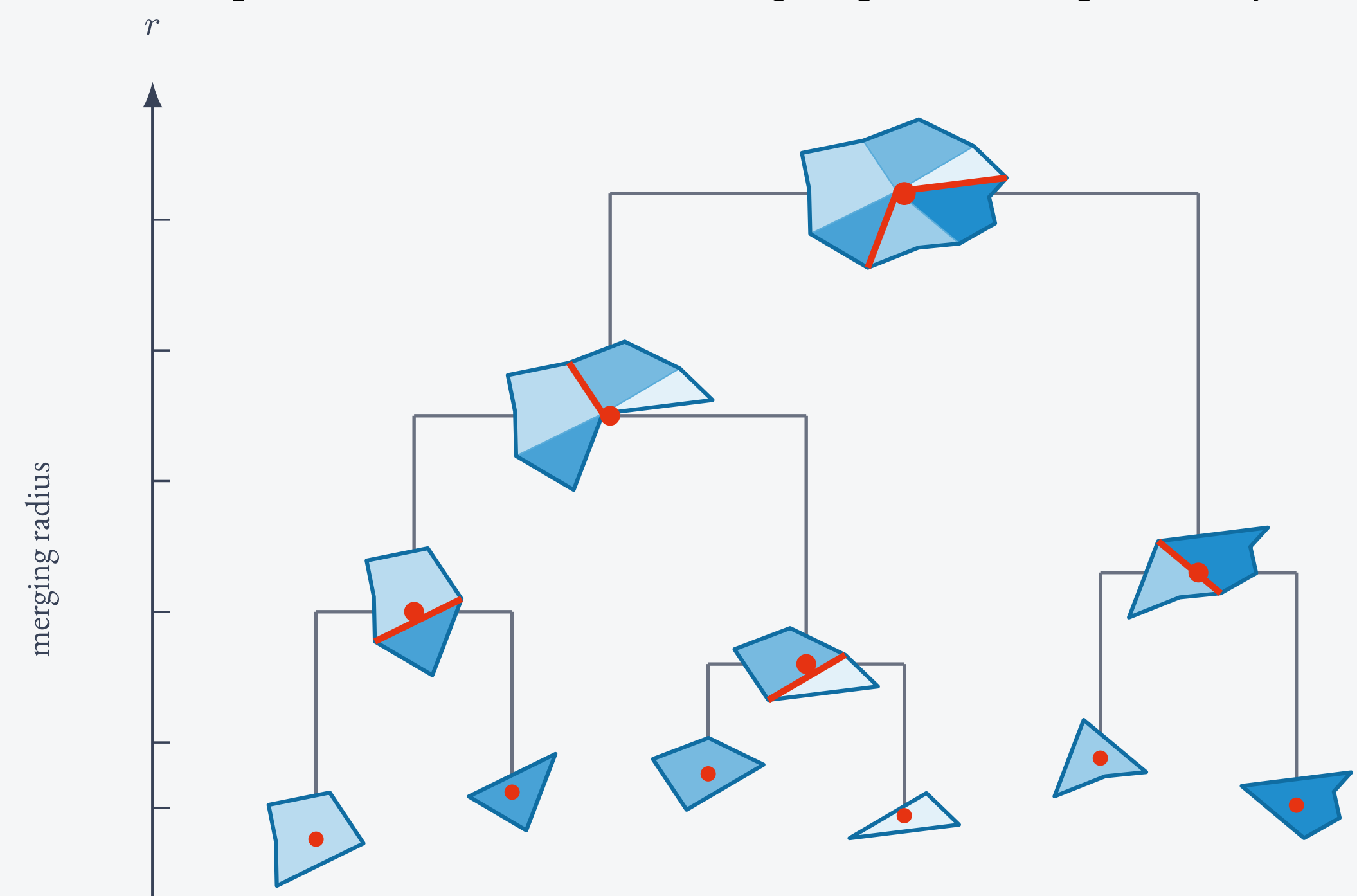
Anomaly
instances

Use model agreement and geometric priors to select branches of the HGP hierarchy. A bounding-box post-processing step reunites fragmented detections [6].

No training is required. The hierarchy becomes an explicit space for geometric reasoning.

5 Perspective: guide a 3D foundation model

Polyhedra as tokens; the hierarchy as multiscale context. Use edges, triangles and tetrahedra to build geometric components, rather than treating all points independently.



Polyhedral
components

Vector tokens
 z_v

Hierarchy-guided
3D backbone

Encode each component's geometry and point features into a vector. Use ancestry and merging scales to guide attention across scales.

Research hypothesis: this geometric structure may improve learned 3D representations. It remains to be tested against the same backbone without hierarchical guidance.

6 From research to Percolia

PERCOLIA

A project led by
Louis HAUSEUX & Alban HAUSEUX,
entering Inria Startup Studio.

Towards geometry-aware tools
for 3D perception.

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Project & poster